Physician suicide: a fleeting moment of despair.

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This ongoing column is dedicated to the challenging clinical interface between psychiatry and primary care-two fields that are inexorably linked.In this article, we examine the somber issue of suicide among physicians. While there is some variation in the literature regarding prevalence, the majority of studies indicates higher rates of suicide among physicians than the general population, particularly among female physicians. Potential contributory factors include various Axis I disorders (especially mood, drug use, and alcohol-related disorders), cognitive style, psychosocial factors, and particular personality characteristics. We urge physicians to be aware of these risk factors.

It's okay to talk: suicide awareness simulation.

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Doctors are at an increased risk of suicide compared with the general population, and there is a current lack of formal education on suicide prevention for peers and colleagues. This educational project aimed to increase suicide awareness for medical students through simulation. A simulation scenario was designed centred around a junior doctor (a qualified doctor who has not yet completed specialist postgraduate training) disclosing thoughts of suicide. The scenario and debriefing were designed using learning objectives and constructive alignment theory. Senior medical students participated in the scenario, which was followed by a facilitated debriefing and the provision of a framework for discussing suicide with a colleague. Quantitative and qualitative student feedback was collected and analysed. A simulation scenario was designed centred around a junior doctor colleague found distressed at work and disclosing thoughts of suicide RESULTS: A total of 35 students participated in the simulation over six sessions. Feedback indicated that students felt this subject was important and that the learning objectives had been achieved. This simulation scenario focusing on suicide awareness for senior medical students has provided opportunity for open discussion and reflection on the topic and has increased the awareness and understanding of suicidality in colleagues. This is one step in the direction of preventing further deaths by suicide in health professionals.

Mental illness and suicide among physicians.

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The COVID-19 pandemic has heightened interest in how physician mental health can be protected and optimised, but uncertainty and misinformation remain about some key issues. In this Review, we discuss the current literature, which shows that despite what might be inferred during training, physicians are not immune to mental illness, with between a quarter and a third reporting increased symptoms of mental ill health. Physicians, particularly female physicians, are at an increased risk of suicide. An emerging consensus exists that some aspects of physician training, working conditions, and organisational support are unacceptable. Changes in medical training and health systems, and the additional strain of working through a pandemic, might have amplified these problems. A new evidence-informed framework for how individual and organisational interventions can be used in an integrated manner in medical schools, in health-care settings, and by professional colleagues is proposed. New initiatives are required at each of these levels, with an urgent need for organisational-level interventions, to better protect the mental health and wellbeing of physicians.

Suicide in physicians and veterinarians: risk factors and theories.

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Physicians and veterinarians are at increased risk for suicide compared to the general population. In particular, this risk appears to be especially pertinent to females in both of these professions. Although increased risk is well-documented, less is known about potential causes for suicidality in these groups. A host of risk factors have been examined in recent research, including job stressors, personality traits, access to lethal medications, and unique work experiences. In addition to these factors, the interpersonal psychological theory of suicidal behavior may provide promise in specifying why physicians and veterinarians are at increased risk for suicide. While there is recognition of mental health issues in these professions, significant treatment barriers remain.

Suicide mortality among physicians, dentists, veterinarians, and pharmacists as well as other high-skilled occupations in Austria from 1986 through 2020.

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Some evidence puts health professionals at increased risk of suicide, especially females, whereas other research suggests a lower risk in high-skilled occupations. This study investigated the suicide risk of four health professions (physicians, dentists, veterinarians, pharmacists) and three other high-skilled occupations (notaries, lawyers, tax advisors/public accountants) in Austria compared to the general population, and analyzed suicide methods across occupations. Data was collected from professional associations and Austrian cause-of-death statistics to determine suicide cases. Gender-specific standardized mortality ratios (SMRs), crude and age-adjusted suicide rates and frequencies for suicide methods were calculated for each profession (maximum time span 1986-2020). Among males, only veterinarians had a significantly elevated suicide risk compared to the general population. Physicians and tax advisors/public accountants had a significantly lower suicide risk. Among females, the veterinarians, physicians, and pharmacists had a significantly elevated suicide risk; for dentists, it was also elevated, though non-significantly. Age-adjusted suicide rates showed a smaller gap between men and women in all professions compared to the general population. Poisoning was the predominant suicide method among health professions, except dentists. These findings are consistent with some of the prior literature and call for specific suicide prevention efforts in health professions, focusing on women.

Can dead man tooth do tell tales? Tooth prints in forensic identification.

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We know that teeth trouble us a lot when we are alive, but they last longer for thousands of years even after we are dead. Teeth being the strongest and resistant structure are the most significant tool in forensic investigations. Patterns of enamel rod end on the tooth surface are known as tooth prints. This study is aimed to know whether these tooth prints can become a forensic tool in personal identification such as finger prints. A study has been targeted toward the same. In the present in-vivo study, acetate peel technique has been used to obtain the replica of enamel rod end patterns. Tooth prints of upper first premolars were recorded from 80 individuals after acid etching using cellulose acetate strips. Then, digital images of the tooth prints obtained at two different intervals were subjected to biometric conversion using Verifinger standard software development kit version 6.5 software followed by the use of Automated Fingerprint Identification System (AFIS) software for comparison of the tooth prints. Similarly, each individual's finger prints were also recorded and were subjected to the same software. Further, recordings of AFIS scores obtained from images were statistically analyzed using Cronbach's test. We observed that comparing two tooth prints taken from an individual at two intervals exhibited similarity in many cases, with wavy pattern tooth print being the predominant type. However, the same prints showed dissimilarity when compared with other individuals. We also found that most of the individuals with whorl pattern finger print showed wavy pattern tooth print and few loop type fingerprints showed linear pattern of tooth prints. Further more experiments on both tooth prints and finger prints are required in establishing an individual's identity.

Reliability of ameloglyphics in forensic identification: a systematic review.

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Teeth are considered as hard tissue analogue to fingerprints, being unique to an individual. The enamel which forms the outer layer of the tooth is formed through a highly dynamic process in which ameloblasts lay down enamel rods in an undulating and intertwining path, which is reflected as a series of enamel rod pattern. The study of these patterns is termed as "Ameloglyphics". These patterns formed on the surface of enamel are called tooth prints each of which comprises the combination of different sub patterns, and are unique for every single tooth of an individual. This characteristic of uniqueness of the tooth print can serve as a significant biometric tool in forensic identification. To assess if ameloglyphics is an accurate and reliable method for forensic identification of individuals. Registration number- CRD42022338138. Data sources- Google Scholar, PubMed, Proquest, Europe PMC, Scopus. Study appraisal and synthesis methods- Studies were assessed for quality with the help of predetermined criteria which categorized the studies into high, medium and low quality with the help of JBI critical appraisal checklist for analytical cross-sectional studies using the key words such as Ameloglyphics, tooth prints and enamel rod end patterns. A total of 475 articles were obtained in the primary search from PubMed, Scopus, Europe PMC, ProQuest, and Google Scholar including gray literature for thorough search of related publications on Ameloglyphics. Following initial search of titles, 433 articles were excluded because they were not related to the objectives of the systematic review and 42 articles were further excluded after abstract reading and removal of duplicates. Finally,18 out of 22 articles which were fulfilling the inclusion criteria of the study were selected for qualitative synthesis after full-text screening and exclusion of review articles. The data obtained from the eighteen included studies were systematically reviewed but a meta-analysis could not be performed due to heterogeneity of the data. Ameloglyphics is a relatively newer technique and a fair number of research are done on its application in identification of individuals. The results of the eighteen studies included in this review suggest that Ameloglyphicsis an useful adjunct for forensic identification. The study protocol can be accessed through the International Prospective Register of Systematic Reviews, the PROSPERO database with the following register number: CRD42022338138.

Forensic Odontology, a Boon and a Humanitarian Tool: A Literature Review.

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Our society faces fresh challenges in every conceivable way, from an increase in the crime rate to a rise in natural disasters. In spite of the leaps in modern technology, medical breakthroughs in the identification of criminals remain a cumbersome task. The modern-day criminal investigation typically involves many different disciplines to solve a crime. The identification of a person's body is required when it is completely mutilated, disfigured, or beyond recognition -- which might be due to any barbaric crime, accident, war, fire, or any natural disaster. Dental identification is one of the most reliable methods, as teeth and dental structures may survive adverse conditions. The techniques involved in forensic odontology include: bite mark analysis, tooth prints, rugroscopy, cheiloscopy, dental DNA analysis, radiographs, and photographs.

Scanning Electron Microscope Corroboration of Ameloglyphics - A New Tool in Forensic Odontology.

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Human teeth resist decomposition to the maximum and has immense potential to serve as hard-tissue counterpart to dermatoglyphics in forensic odontology. Ameloglyphics is the science of recording and analyzing the tooth print. To assess the scope of viability, reproducibility, and identification of enamel prints (akin to fingerprints) and their patterns as a tool for identification. To establish that expression of enamel prints is a direct result of the enamel rod configuration on the surface of the crown as detected by scanning electron microscope (SEM). The teeth samples ($n = 10$) were first analyzed through (SEM) and the image of the arrangement of rods on the surface was captured. Enamel prints were registered in a standard procedure by virtue of ink transfer on a cellophane tape from etched tooth enamel surface of the same samples. These prints and SEM images were subjected to Rapid Sizer® image editing software to obtain a pattern (sketched outline image software). Patterns were identified manually. Reproducibility, specificity, and feasibility of the above procedure were determined. There appeared to be a high rate of reproducibility (98%-100%) and specificity (100%). The paraphernalia required as well as the technique entrenched were feasible. Furthermore, the SEM analysis established the viability and reliability. Ameloglyphics is a sensitive and reproducible scientific tool that can be utilized for the management, examination, and evaluation of dental evidence for identification at crime scene and disaster sites. Its importance vis-a-vis fingerprints cannot be understated, especially in view of the seeming indestructibility of the enamel.

Analysis of enamel rod end pattern for personal identification.

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Microscopically, groups of enamel rods run in unique direction, which differ from adjacent group of enamel rods and results in forming different patterns of enamel rod endings on tooth surface. These are called as tooth prints and they help in personal identification in forensic odontology. The aim of the present study is to analyze the enamel rod end pattern on the tooth surface for personal identification and to analyze the familial inheritance of enamel rod end pattern. In the present study, 100 different families were considered for the analysis of tooth print pattern. In each family, four members were present. The maxillary central incisor, canine and first premolar were selected. Enamel rod end pattern was recorded using acetate peel technique and analyzed using Verifinger® standard SDK version 6.7 software. Data analysis was performed using the SPSS software. Contingency coefficient statistical analysis was used for the comparison of tooth print pattern in incisors, canines and premolars based on age and gender. $P < 0.05$ was considered statistically significant. The present study showed that a tooth print is composed of combination of eight distinct subpatterns, namely wavy branched, wavy unbranched, linear branched, linear unbranched, whorl open, whorl closed, loop and stem-like pattern. Wavy branched pattern was found to be the most predominant pattern in incisors, canines and first premolars in our study. Familial tendency of tooth print pattern in incisors, canines and premolars was noticed in 65%, 66% and 52% of the families, respectively. Tooth prints are unique to an individual and can be used as a valuable inexpensive tool in forensic odontology for personal identification.